NPTEL Online Certification Course

Course: Cloud Computing & Distributed Systems | Week 3: Virtualization & Hypervisors

1. Introduction to Virtualization in Cloud Computing

Virtualization is the fundamental foundational technology that enables cloud computing. It allows multiple operating systems and applications to run concurrently on a single physical host by abstracting the underlying physical hardware resources. Through virtualization, cloud service providers achieve high resource utilization, multi-tenancy, and rapid elasticity.

Essential goals of virtualization in cloud infrastructure include:

- **Resource Pooling:** Aggregating compute, memory, and storage across homogeneous or heterogeneous servers.
- **Isolation & Security:** Ensuring guest virtual machines cannot access or compromise each other's memory space.
- **Workload Migration:** Enabling live migration of running VMs between physical hosts with minimal downtime.
- **High Availability & Disaster Recovery:** Rapid checkpointing and snapshot restoration.

2. Hypervisor Taxonomy: Type-1 vs Type-2

A Hypervisor, also termed a Virtual Machine Monitor (VMM), is the software layer that controls hardware access for guest VMs. Hypervisors are strictly classified into two primary architectural types:

Attribute	Type-1 Hypervisor (Bare-Metal)	Type-2 Hypervisor (Hosted)
Underlying Layer	Directly on Physical Hardware	Runs on top of a Host Operating System
Performance	Near-native performance, low latency	Lower performance due to OS context switches
Resource Overhead	Minimal microkernel footprint	Significant footprint of host OS + hypervisor
Typical Use Cases	Enterprise Datacenters, AWS EC2, Azure	Software testing, Local development
Popular Examples	VMware ESXi, Xen, KVM, MS Hyper-V	Oracle VirtualBox, VMware Workstation

3. Full Virtualization vs Paravirtualization

In classical x86 architecture, 17 sensitive instructions could not be virtualized without special intervention. Two prominent software techniques evolved to solve this:

- **Full Virtualization (Binary Translation):** Guest OS is completely unmodified. Sensitive privileged instructions are dynamically rewritten at runtime into safe hypervisor instructions. Examples include early VMware ESX.
- **Paravirtualization (Hypercalls):** The guest OS kernel is modified to recognize it is running in a virtualized container. Instead of executing sensitive instructions, it makes direct software traps called *Hypercalls* to the VMM (pioneered by Xen).
- **Hardware-Assisted Virtualization:** Intel VT-x and AMD-V added root/non-root CPU modes (Ring -1), allowing unmodified guest OS to run at native speeds with hardware-enforced traps.

4. Service Level Agreements (SLA) & Reliability Calculations

In commercial cloud deployments, availability is governed by legal Service Level Agreements (SLAs). System availability A is formulated mathematically as:

Availability Formula: A = MTTF / (MTTF + MTTR) * 100%

Where **MTTF** is Mean Time To Failure, and **MTTR** is Mean Time To Repair.

- Five Nines (99.999%): Maximum allowed downtime of ~5.26 minutes per year.
- Four Nines (99.99%): Maximum allowed downtime of ~52.6 minutes per year.
- Three Nines (99.9%): Maximum allowed downtime of ~8.76 hours per year.

5. NPTEL Assignment Review & Key Questions

Be prepared to calculate serial and parallel availability and distinguish between elasticity (autonomic workload adaptation) and scalability (structural capacity expansion).